

Osaid Khan

[+1 \(945\) 304-5781](tel:+19453045781) | khanosaid726@gmail.com | www.linkedin.com/in/-osaid-khan/ | github.com/ozzyozbourne

Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the **System Development Engineer** position with **Amazon Robotics**. I bring over three years of professional experience in infrastructure automation, platform integration, and scalable system development, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University, NY.

Through my experience across two engineering roles, I have built and delivered production-grade systems that directly reflect the technical depth and operational excellence this position demands. As a Software Engineer at Sperse, I architected a platform integration layer that automates 1,000+ endpoint workflows with safe, validated execution — requiring rigorous workflow design, business rule implementation, and reliable execution across a distributed platform surface with minimal human intervention. To keep these systems production-ready, I built a comprehensive observability and monitoring layer using Langfuse and LangSmith, covering tracing, structured logs, and evaluation hooks to proactively troubleshoot and resolve issues, and implemented performance optimization strategies that reduced latency by 60% and API costs by 50%, directly enhancing system responsiveness and scalability. Prior to Sperse, as a Software Engineer at Qualitest, I contributed to infrastructure that thousands of users rely on daily. I designed a cloud-agnostic Storage Service using Python, PostgreSQL, and Redis that seamlessly handled file operations across AWS S3, Google Cloud Storage, and MinIO — giving me direct, hands-on experience with multi-cloud infrastructure automation and scalable system design. I also contributed to an Identity & Access Management platform built with React, Spring Boot, and AWS, supporting thousands of users across 30+ applications, and developed automation workflows using AWS Step Functions to handle long-running data import jobs and prevent Lambda timeouts — both of which required the kind of systems-level thinking, operational stability focus, and commitment to operational excellence this role demands. Together, these roles have prepared me to contribute meaningfully to Amazon Robotics' supply chain planning systems and drive impact within a cross-functional engineering team.

My academic research experience has given me a rigorous foundation in high-performance computing, infrastructure optimization, and technical communication — skills that translate directly into the engineering discipline and collaborative culture Amazon values. As an AI Researcher at the Pace Artificial Intelligence Lab, I conducted applied research on StyleGAN and Stable Diffusion, leveraging CUDA, PyTorch, and LoRA on Pace University's HPC cluster for model training and fine-tuning, optimizing for high-performance at global scale. I was responsible for managing the full infrastructure lifecycle — from training runs to deploying inference infrastructure using vLLM and Hugging Face, ensuring secure, reliable, and efficient serving with minimal operational overhead. Beyond building the systems myself, I developed training materials and conducted workshops on GPU programming and building AI Agents for an audience of 50+ students, providing hands-on support and documentation. Designing and delivering those sessions required me to break down complex technical concepts clearly and accessibly — directly mirroring the training, support, and knowledge-sharing responsibilities central to this role. This experience has shaped me into an engineer who builds with rigor and communicates with clarity, both of which I would bring to Amazon Robotics and its mission.

I have always been passionate about customer obsession and commitment to operational excellence, and envision myself contributing my talent to the furthering of Amazon's mission to be Earth's most customer-centric company. I would love to discuss my interest in this position with your team and can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration, I look forward to connecting!

Sincerely,
Osaid Khan